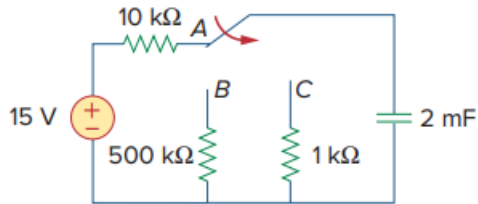


Problem

Assuming that the switch in Fig. 7.87 has been in position *A* for a long time and is moved to position *B* at $t = 0$. Then at $t = 1$ second, the switch moves from *B* to *C*. Find $v_C(t)$ for $t \geq 0$.

Fig. 7.87:

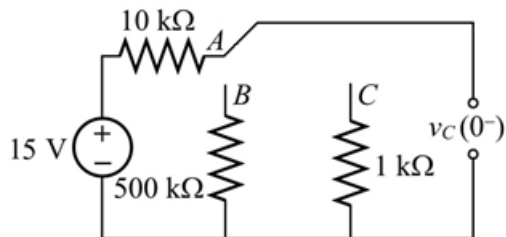


Step-by-step solution

Step 1 of 5

Refer to Figure 7.87 in the text book for the circuit.

The switch is in position *A* for a long time. Under steady state condition capacitor acts as an open circuit. Draw the circuit for $t < 0$.



The capacitor voltage at $t = 0^-$ is,

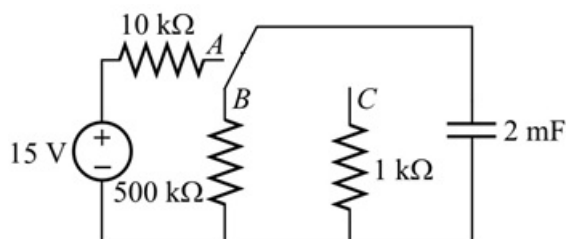
$$v_C(0^-) = 15 \text{ V}$$

The voltage across the capacitor cannot change instantaneously.

$$\begin{aligned} v_C(0^-) &= v_C(0^+) \\ &= v_C(0) \\ &= 15 \text{ V} \end{aligned}$$

Step 2 of 5

The switch moves to position *B* at $t = 0$. Draw the circuit for $t > 0$ when the switch is in position *B*.



The circuit is a source-free *RC* circuit.

Step 3 of 5

Determine the time constant, τ of the RC circuit.

$$\begin{aligned}\tau &= RC \\ &= (500 \text{ k})(2 \text{ m}) \\ &= 1000\end{aligned}$$

Determine the capacitor voltage of the circuit.

$$v_C(t) = v_C(0)e^{-\frac{t}{\tau}}$$

Substitute 15 V for $v_C(0)$ and 1000 for τ .

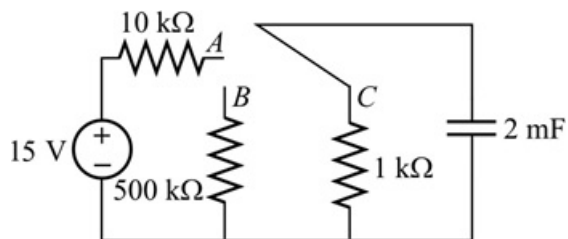
$$v_C(t) = 15e^{-\frac{t}{1000}} \text{ V}$$

Step 4 of 5

Determine the capacitor voltage at $t = 1 \text{ s}$.

$$\begin{aligned}v_C(1) &= 15e^{-\frac{1}{1000}} \\ &= 14.985 \text{ V}\end{aligned}$$

The switch moves to position C at $t = 1 \text{ s}$. Draw the circuit for $t > 1 \text{ s}$.



The circuit is a source-free RC circuit.

Step 5 of 5

Determine the time constant, τ of the RC circuit.

$$\begin{aligned}\tau &= RC \\ &= (1 \text{ k})(2 \text{ m}) \\ &= 2\end{aligned}$$

Determine the capacitor voltage of the circuit.

$$v_C(t) = v_C(1)e^{-\frac{t-1}{\tau}}$$

Substitute 14.985 V for $v_C(1)$ and 2 for τ .

$$\begin{aligned}v_C(t) &= 14.985e^{-\frac{t-1}{2}} \\ &= 14.985e^{-0.5(t-1)} \text{ V}\end{aligned}$$

Thus, the capacitor voltage $v_C(t)$ for $t \geq 0$ is,

$\begin{aligned}v_C(t) &= 15e^{-\frac{t}{1000}} \text{ V} && ; 0 \leq t \leq 1 \text{ s} \\ &= 14.985e^{-0.5(t-1)} \text{ V} && ; t > 1 \text{ s}\end{aligned}$
